

Tenancy and Organization Foundation

Phase 2 Architecture Report

Exact MySQL 8.4 schema, Laravel domain foundation, organization hierarchy, ERD, access-point masters and implementation controls.

Version 0.2 · 20 July 2026 · Draft for Technical Review

Canonical target: Laravel 13 + PHP 8.3+ + MySQL 8.4 LTS

RK3588 Android terminal remains the sole intelligent edge controller.

Outcome and architecture decisions

Phase 2 creates the ownership spine used by every later employee, device, attendance, access, leave and payroll record.

17

organization and tenancy tables

143

named keys and constraints

17

Eloquent domain models

Binding decisions

1

Hybrid tenancy

Shared tenant_id database by default; dedicated databases remain a deployment option.

2

Stable identity

BIGINT UNSIGNED internal keys plus ULID public_id values for public references.

3

Database-enforced scope

Composite foreign keys prove tenant, company, plant, branch and location consistency.

4

Restrictive lifecycle

RESTRICT protects referenced history; selected master rows use soft deletion.

5

Time discipline

UTC is persisted; IANA timezones are retained only for local interpretation.

6

Code discipline

No database ENUMs; stable values use PHP backed enums and request validation.

Explicit boundary: Phase 2 adds no employees, credentials, devices, relays, attendance events or payroll rules. Those domains attach to this foundation in later phases.

Master logical ERD

The model separates corporate masters, physical sites, workforce catalogues and access-point masters while retaining one tenant/company ownership spine.



Parent	Primary children	Guardrail
Tenant	Companies	Trusted middleware resolves tenant context.
Company	Legal entities, business units, plants, branches, departments, grades	Composite keys reject cross-company references.
Plant / Branch	Locations; optional department and cost-centre scope	A location belongs to exactly one site kind.
Location	Buildings, zones, gates and doors	Physical children carry location-aware foreign keys.
Grade / Gate	Designations / Doors	Both remain reusable masters, not device configuration.

Physical organization hierarchy

The physical hierarchy supports industrial plants, offices, depots and mixed campuses without forcing a single rigid building model.

Site ownership path



Physical containment



Flexible placement

- Zones may scope a whole location, building or floor.
- Gates identify checkpoints at a location and can reference a zone.
- Doors can reference a gate plus optional building, floor and zone context.
- Basement floors use negative level numbers.

Validation invariants

- A location cannot belong to both a plant and a branch.
- Coordinates are validated as a pair and kept at WGS84 precision.
- Physical references must resolve inside the same tenant/company/location.
- Door fail mode changes require privileged approval and audit reason.

Gates and doors are structural masters only. Terminal, credential-reader, relay and lock-channel assignments are intentionally deferred to the device phases.

Table catalogue

Every table is included in the executable migration, raw MySQL DDL, model catalogue and detailed specification.

Table	Purpose	Deletion	Volume
tenants	Top-level customer boundary and data-isolation configuration.	Soft + RESTRICT	Low
companies	Operational company master inside a tenant; the main reporting and ownership spine.	Soft + RESTRICT	Low
legal_entities	Registered legal bodies used for statutory, tax, payroll and contract identity.	Soft + RESTRICT	Low
business_units	Flexible commercial or operating hierarchy inside a company.	Soft + RESTRICT	Low
plants	Manufacturing or industrial operating units.	Soft + RESTRICT	Low
branches	Non-plant offices, depots, service centres or administrative branches.	Soft + RESTRICT	Low
locations	Addressable sites where attendance, geofencing or access control can operate.	Soft + RESTRICT	Low
buildings	Named structures within a location.	Soft + RESTRICT	Low
floors	Ordered levels within a building, including negative basement levels.	Soft + RESTRICT	Low
zones	Logical or physical subdivisions used for restricted access and later occupancy rules.	Soft + RESTRICT	Low
departments	Functional organization tree with optional plant, branch and location context.	Soft + RESTRICT	Low
department_hierarchy	Closure table that makes ancestor and descendant department queries fast and cycle-safe.	Service-managed	Medium
cost_centres	Finance and payroll allocation codes attached to organization nodes.	Soft + RESTRICT	Low
grades	Company-defined employee grade or band catalogue.	Soft + RESTRICT	Low
designations	Company-defined job titles, optionally grouped by grade.	Soft + RESTRICT	Low
gates	Named entry, exit or bidirectional checkpoints at a location.	Soft + RESTRICT	Low
doors	Physical controlled openings; device and relay assignments are added in later phases.	Soft + RESTRICT	Low

tenants

Top-level customer boundary and data-isolation configuration.

TABLE 01

17 columns

Tenant model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK	Internal BIGINT surrogate primary key; never expose it in public APIs.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
slug	VARCHAR(160)	Y	-	UNIQUE	Domain attribute documented and validated by the matching Form Request and service.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
isolation_mode	VARCHAR(24)	Y	'shared'	-	Domain attribute documented and validated by the matching Form Request and service.
default_timezone	VARCHAR(64)	Y	'UTC'	-	IANA timezone name used only for local interpretation; persisted event time remains UTC.
default_locale	VARCHAR(12)	Y	'en'	-	Domain attribute documented and validated by the matching Form Request and service.
country_code	CHAR(2)	N	-	-	External or domain code used for integration and reporting.
data_residency_region	VARCHAR(64)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
activated_at	TIMESTAMP(6)	N	-	-	UTC event or lifecycle timestamp.
suspended_at	TIMESTAMP(6)	N	-	-	UTC event or lifecycle timestamp.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	INDEX	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- Root table; no parent foreign key in Phase 2.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: TenantController

Service: TenantProvisioningService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=TNT-001; name=Demo Tenants; status=active; isolation_mode=shared; default_timezone=UTC
```

companies

Operational company master inside a tenant; the main reporting and ownership spine.

TABLE 02

22 columns

Comanie model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
legal_name	VARCHAR(200)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
trade_name	VARCHAR(160)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
slug	VARCHAR(160)	Y	-	UNIQUE	Domain attribute documented and validated by the matching Form Request and service.
email	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
phone	VARCHAR(32)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
website	VARCHAR(255)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
timezone	VARCHAR(64)	Y	'UTC'	-	IANA timezone name used only for local interpretation; persisted event time remains UTC.
default_currency	CHAR(3)	Y	'INR'	-	Domain attribute documented and validated by the matching Form Request and service.
country_code	CHAR(2)	Y	'IN'	-	External or domain code used for integration and reporting.
fiscal_year_start_month	TINYINT UNSIGNED	Y	4	-	Domain attribute documented and validated by the matching Form Request and service.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	INDEX	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: CompanyController

Service: CompanyStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=CMP-001; name=Demo Companies; legal_name=Demo Companies; timezone=UTC
```

legal_entities

Registered legal bodies used for statutory, tax, payroll and contract identity.

TABLE 03

23 columns

LegalEntity model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
legal_name	VARCHAR(200)	Y	-	-	Domain attribute documented and validated by the matching Form Request and service.
registration_number	VARCHAR(80)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
tax_identifier	VARCHAR(80)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
email	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
phone	VARCHAR(32)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
address_line_1	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
address_line_2	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
city	VARCHAR(100)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
state_code	VARCHAR(32)	N	-	-	External or domain code used for integration and reporting.
country_code	CHAR(2)	Y	'IN'	-	External or domain code used for integration and reporting.
postal_code	VARCHAR(20)	N	-	-	External or domain code used for integration and reporting.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: LegalEntityController
Service: CompanyStructureService
Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

id=1; tenant_id=1; company_id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=LE-001; legal_name=Demo Legal Entities; country_code=IN

business_units

Flexible commercial or operating hierarchy inside a company.

TABLE 04

16 columns

BusinessUnit model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
parent_business_unit_id	BIGINT UNSIGNED	N	-	INDEX, FK · business_units(tenant_id,company_id,i d)	Foreign key to the selected parent business unit; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
description	TEXT	N	-	-	Optional administrator-facing explanation.
sort_order	SMALLINT UNSIGNED	Y	0	-	Stable display or hierarchy ordering value.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, parent_business_unit_id -> business_units; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: BusinessUnitController

Service: CompanyStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; parent_business_unit_id=None; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=BU-001; name=Demo Business Units
```

plants

Manufacturing or industrial operating units.

TABLE 05

16 columns

Plant model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
legal_entity_id	BIGINT UNSIGNED	N	-	INDEX, FK · legal_entities(tenant_id,company_id,id)	Foreign key to the selected legal entity; service validation also checks company/location scope.
business_unit_id	BIGINT UNSIGNED	N	-	INDEX, FK · business_units(tenant_id,company_id,id)	Foreign key to the selected business unit; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
timezone	VARCHAR(64)	Y	'UTC'	-	IANA timezone name used only for local interpretation; persisted event time remains UTC.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, legal_entity_id -> legal_entities; delete is restricted.
- tenant_id, company_id, business_unit_id -> business_units; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: PlantController

Service: CompanyStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; legal_entity_id=None; business_unit_id=None; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=PLT-001
```

branches

Non-plant offices, depots, service centres or administrative branches.

TABLE 06

16 columns

Branche model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
legal_entity_id	BIGINT UNSIGNED	N	-	INDEX, FK · legal_entities(tenant_id,company_id,id)	Foreign key to the selected legal entity; service validation also checks company/location scope.
business_unit_id	BIGINT UNSIGNED	N	-	INDEX, FK · business_units(tenant_id,company_id,id)	Foreign key to the selected business unit; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
timezone	VARCHAR(64)	Y	'UTC'	-	IANA timezone name used only for local interpretation; persisted event time remains UTC.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, legal_entity_id -> legal_entities; delete is restricted.
- tenant_id, company_id, business_unit_id -> business_units; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: BranchController

Service: CompanyStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; legal_entity_id=None; business_unit_id=None; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=BR-001
```

locations

Addressable sites where attendance, geofencing or access control can operate.

TABLE 07

30 columns

Location model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
business_unit_id	BIGINT UNSIGNED	N	-	FK · business_units(tenant_id,company_id,id)	Foreign key to the selected business unit; service validation also checks company/location scope.
plant_id	BIGINT UNSIGNED	N	-	INDEX, FK · plants(tenant_id,company_id,id)	Foreign key to the selected plant; service validation also checks company/location scope.
branch_id	BIGINT UNSIGNED	N	-	INDEX, FK · branches(tenant_id,company_id,id)	Foreign key to the selected branch; service validation also checks company/location scope.
parent_location_id	BIGINT UNSIGNED	N	-	INDEX, FK · locations(tenant_id,company_id,id)	Foreign key to the selected parent location; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
location_kind	VARCHAR(32)	Y	'site'	-	Domain attribute documented and validated by the matching Form Request and service.
address_line_1	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
address_line_2	VARCHAR(190)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
city	VARCHAR(100)	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
state_code	VARCHAR(32)	N	-	-	External or domain code used for integration and reporting.
country_code	CHAR(2)	Y	'IN'	-	External or domain code used for integration and reporting.
postal_code	VARCHAR(20)	N	-	-	External or domain code used for integration and reporting.
latitude	DECIMAL(10,7)	N	-	-	WGS84 latitude retained to seven decimal places.
longitude	DECIMAL(10,7)	N	-	-	WGS84 longitude retained to seven decimal places.
default_geofence_radius_m	INT UNSIGNED	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
timezone	VARCHAR(64)	Y	'UTC'	-	IANA timezone name used only for local interpretation; persisted event time remains UTC.
attendance_enabled	TINYINT(1)	Y	1	-	Boolean feature or classification flag used by policy and access-control services.
access_control_enabled	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, business_unit_id -> business_units; delete is restricted.
- tenant_id, company_id, plant_id -> plants; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: LocationController

Service: LocationStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

id=1; tenant_id=1; company_id=1; business_unit_id=None; plant_id=None; branch_id=None; parent_location_id=None

buildings

Named structures within a location.

TABLE 08

15 columns

Building model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
location_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
description	TEXT	N	-	-	Optional administrator-facing explanation.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, location_id -> locations; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: BuildingController

Service: LocationStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

id=1; tenant_id=1; company_id=1; location_id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=BLD-001; name=Demo Buildings

floors

Ordered levels within a building, including negative basement levels.

TABLE 09

15 columns

Floor model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
location_id	BIGINT UNSIGNED	Y	-	UNIQUE, FK · buildings(tenant_id,company_id,location_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
building_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · buildings(tenant_id,company_id,location_id,id)	Foreign key to the selected building; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(120)	Y	-	-	Display name used by administration and reporting.
level_number	SMALLINT	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
sort_order	SMALLINT UNSIGNED	Y	0	INDEX	Stable display or hierarchy ordering value.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, location_id, building_id -> buildings; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: FloorController

Service: LocationStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; location_id=1; building_id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=FLR-001
```

zones

Logical or physical subdivisions used for restricted access and later occupancy rules.

TABLE 10

18 columns

Zone model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
location_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
building_id	BIGINT UNSIGNED	N	-	FK · buildings(tenant_id,company_id,location_id,id)	Foreign key to the selected building; service validation also checks company/location scope.
floor_id	BIGINT UNSIGNED	N	-	FK · floors(tenant_id,company_id,location_id,building_id,id)	Foreign key to the selected floor; service validation also checks company/location scope.
parent_zone_id	BIGINT UNSIGNED	N	-	INDEX, FK · zones(tenant_id,company_id,location_id,id)	Foreign key to the selected parent zone; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(120)	Y	-	-	Display name used by administration and reporting.
zone_type	VARCHAR(32)	Y	'general'	-	Domain attribute documented and validated by the matching Form Request and service.
is_restricted	TINYINT(1)	Y	0	INDEX	Boolean feature or classification flag used by policy and access-control services.
capacity	INT UNSIGNED	N	-	-	Domain attribute documented and validated by the matching Form Request and service.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, location_id -> locations; delete is restricted.
- tenant_id, company_id, location_id, building_id -> buildings; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: ZoneController

Service: LocationStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; location_id=1; building_id=None; floor_id=None; parent_zone_id=None
```

departments

Functional organization tree with optional plant, branch and location context.

TABLE 11

20 columns

Department model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
business_unit_id	BIGINT UNSIGNED	N	-	FK · business_units(tenant_id,company_id,id)	Foreign key to the selected business unit; service validation also checks company/location scope.
plant_id	BIGINT UNSIGNED	N	-	FK · plants(tenant_id,company_id,id)	Foreign key to the selected plant; service validation also checks company/location scope.
branch_id	BIGINT UNSIGNED	N	-	FK · branches(tenant_id,company_id,id)	Foreign key to the selected branch; service validation also checks company/location scope.
location_id	BIGINT UNSIGNED	N	-	INDEX, FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
parent_department_id	BIGINT UNSIGNED	N	-	INDEX, FK · departments(tenant_id,company_id,id)	Foreign key to the selected parent department; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
description	TEXT	N	-	-	Optional administrator-facing explanation.
sort_order	SMALLINT UNSIGNED	Y	0	-	Stable display or hierarchy ordering value.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, business_unit_id -> business_units; delete is restricted.
- tenant_id, company_id, plant_id -> plants; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: DepartmentController

Service: DepartmentHierarchyService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; business_unit_id=None; plant_id=None; branch_id=None; location_id=None
```

department_hierarchy

Closure table that makes ancestor and descendant department queries fast and cycle-safe.

TABLE 12

8 columns

DepartmentHierarchy model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
ancestor_department_id	BIGINT UNSIGNED	Y	-	UNIQUE, FK · departments(tenant_id,company_id,id)	Foreign key to the selected ancestor department; service validation also checks company/location scope.
descendant_department_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · departments(tenant_id,company_id,id)	Foreign key to the selected descendant department; service validation also checks company/location scope.
depth	SMALLINT UNSIGNED	Y	-	INDEX	Closure-table distance: self is 0, direct parent is 1.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, ancestor_department_id -> departments; delete is restricted.
- tenant_id, company_id, descendant_department_id -> departments; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: DepartmentController (internal)

Service: DepartmentHierarchyService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; ancestor_department_id=1; descendant_department_id=1
```

cost_centres

Finance and payroll allocation codes attached to organization nodes.

TABLE 13

20 columns

CostCentre model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
legal_entity_id	BIGINT UNSIGNED	N	-	FK · legal_entities(tenant_id,company_id,id)	Foreign key to the selected legal entity; service validation also checks company/location scope.
business_unit_id	BIGINT UNSIGNED	N	-	FK · business_units(tenant_id,company_id,id)	Foreign key to the selected business unit; service validation also checks company/location scope.
plant_id	BIGINT UNSIGNED	N	-	FK · plants(tenant_id,company_id,id)	Foreign key to the selected plant; service validation also checks company/location scope.
branch_id	BIGINT UNSIGNED	N	-	FK · branches(tenant_id,company_id,id)	Foreign key to the selected branch; service validation also checks company/location scope.
location_id	BIGINT UNSIGNED	N	-	FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
department_id	BIGINT UNSIGNED	N	-	INDEX, FK · departments(tenant_id,company_id,id)	Foreign key to the selected department; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(160)	Y	-	-	Display name used by administration and reporting.
external_payroll_code	VARCHAR(64)	N	-	-	External or domain code used for integration and reporting.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, legal_entity_id -> legal_entities; delete is restricted.
- tenant_id, company_id, business_unit_id -> business_units; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: CostCentreController

Service: CompanyStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; legal_entity_id=None; business_unit_id=None; plant_id=None; branch_id=None
```

grades

Company-defined employee grade or band catalogue.

TABLE 14

14 columns

Grade model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(120)	Y	-	-	Display name used by administration and reporting.
rank_order	SMALLINT UNSIGNED	Y	0	INDEX	Stable display or hierarchy ordering value.
description	TEXT	N	-	-	Optional administrator-facing explanation.
status	VARCHAR(24)	Y	'active'	-	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: GradeController

Service: WorkforceCatalogueService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=GR-001; name=Demo Grades; rank_order=0
```

designations

Company-defined job titles, optionally grouped by grade.

TABLE 15

15 columns

Designation model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
grade_id	BIGINT UNSIGNED	N	-	INDEX, FK · grades(tenant_id,company_id,id)	Foreign key to the selected grade; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
title	VARCHAR(160)	Y	-	-	Domain attribute documented and validated by the matching Form Request and service.
description	TEXT	N	-	-	Optional administrator-facing explanation.
is_managerial	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
effective_from	DATE	N	-	-	First business date on which the master record applies.
effective_to	DATE	N	-	-	Last business date on which the master record applies; null means open-ended.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, grade_id -> grades; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: DesignationController

Service: WorkforceCatalogueService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

```
id=1; tenant_id=1; company_id=1; grade_id=None; public_id=01K0A2B3C4D5E6F7G8H9J0K1M2; code=DSG-001; title=Demo Designations
```

gates

Named entry, exit or bidirectional checkpoints at a location.

TABLE 16

20 columns

Gate model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
location_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
building_id	BIGINT UNSIGNED	N	-	FK · buildings(tenant_id,company_id,location_id,id)	Foreign key to the selected building; service validation also checks company/location scope.
floor_id	BIGINT UNSIGNED	N	-	FK · floors(tenant_id,company_id,location_id,building_id,id)	Foreign key to the selected floor; service validation also checks company/location scope.
zone_id	BIGINT UNSIGNED	N	-	FK · zones(tenant_id,company_id,location_id,building_id,id)	Foreign key to the selected zone; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(120)	Y	-	-	Display name used by administration and reporting.
gate_type	VARCHAR(32)	Y	'pedestrian'	-	Domain attribute documented and validated by the matching Form Request and service.
direction_mode	VARCHAR(24)	Y	'bidirectional'	-	Entry, exit or bidirectional behavior; validated by a PHP backed enum.
attendance_enabled	TINYINT(1)	Y	1	-	Boolean feature or classification flag used by policy and access-control services.
access_control_enabled	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
is_emergency_exit	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, location_id -> locations; delete is restricted.
- tenant_id, company_id, location_id, building_id -> buildings; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: GateController**Service:** AccessPointStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

id=1; tenant_id=1; company_id=1; location_id=1; building_id=None; floor_id=None; zone_id=None

doors

Physical controlled openings; device and relay assignments are added in later phases.

TABLE 17

23 columns

Door model

MySQL 8.4

ULID route binding

Column	Type	Req.	Default	Key / FK	Description
id	BIGINT UNSIGNED	Y	-	PK, UNIQUE	Internal BIGINT surrogate primary key; never expose it in public APIs.
tenant_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · tenants(id)	Tenant ownership boundary used by authorization and tenant-aware indexes.
company_id	BIGINT UNSIGNED	Y	-	UNIQUE, INDEX, FK · companies(tenant_id,id)	Owning company; composite foreign keys also prove tenant consistency.
location_id	BIGINT UNSIGNED	Y	-	UNIQUE, FK · locations(tenant_id,company_id,id)	Foreign key to the selected location; service validation also checks company/location scope.
gate_id	BIGINT UNSIGNED	N	-	INDEX, FK · gates(tenant_id,company_id,location_id,id)	Foreign key to the selected gate; service validation also checks company/location scope.
building_id	BIGINT UNSIGNED	N	-	FK · buildings(tenant_id,company_id,location_id,id)	Foreign key to the selected building; service validation also checks company/location scope.
floor_id	BIGINT UNSIGNED	N	-	FK · floors(tenant_id,company_id,location_id,building_id,id)	Foreign key to the selected floor; service validation also checks company/location scope.
zone_id	BIGINT UNSIGNED	N	-	INDEX, FK · zones(tenant_id,company_id,location_id,id)	Foreign key to the selected zone; service validation also checks company/location scope.
public_id	CHAR(26)	Y	-	UNIQUE	Stable ULID used in URLs, APIs, logs and distributed references.
code	VARCHAR(32)	Y	-	UNIQUE	Human-managed short business code, unique within its documented scope.
name	VARCHAR(120)	Y	-	-	Display name used by administration and reporting.
door_type	VARCHAR(32)	Y	'pedestrian'	-	Domain attribute documented and validated by the matching Form Request and service.
direction_mode	VARCHAR(24)	Y	'bidirectional'	-	Entry, exit or bidirectional behavior; validated by a PHP backed enum.
fail_mode	VARCHAR(24)	Y	'fail_secure'	-	Fail-safe or fail-secure door behavior; a safety decision, not a casual UI setting.
normally_locked	TINYINT(1)	Y	1	-	Boolean feature or classification flag used by policy and access-control services.
requires_liveness	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
anti_passback_enabled	TINYINT(1)	Y	0	-	Boolean feature or classification flag used by policy and access-control services.
held_open_timeout_seconds	SMALLINT UNSIGNED	Y	30	-	Domain attribute documented and validated by the matching Form Request and service.
status	VARCHAR(24)	Y	'active'	INDEX	Stable lifecycle code validated by the RecordStatus PHP enum.
settings	JSON	N	-	-	Optional non-critical extension settings; core relationships never live only in JSON.
created_at	TIMESTAMP(6)	N	-	-	UTC creation timestamp managed by Laravel.
updated_at	TIMESTAMP(6)	N	-	-	UTC last-update timestamp managed by Laravel.
deleted_at	TIMESTAMP(6)	N	-	-	Soft-deletion timestamp; critical history remains recoverable.

Relationships and lifecycle

- tenant_id -> tenants; delete is restricted.
- tenant_id, company_id -> companies; delete is restricted.
- tenant_id, company_id, location_id -> locations; delete is restricted.
- tenant_id, company_id, location_id, gate_id -> gates; delete is restricted.
- Critical references use RESTRICT; normal list endpoints exclude soft-deleted rows.

Laravel boundary

Controller: DoorController**Service:** AccessPointStructureService

Policies and Form Requests authorize; the service owns cross-scope validation and transactions.

id=1; tenant_id=1; company_id=1; location_id=1; gate_id=None; building_id=None; floor_id=None

Laravel controller-service-model design

Controllers remain thin. Trusted context, policies and Form Requests reject invalid scope before a domain service opens a transaction.

1 · Context

Authentication middleware resolves tenant, company and actor context.

2 · Request

Policy + Form Request validate ability, format and stable enum values.

3 · Controller

Maps request to a DTO and returns an API resource.

4 · Service

Checks cross-scope invariants and owns the transaction boundary.

5 · Model / DB

Eloquent relations plus composite FKs provide the final integrity barrier.

Representative implementation

- DepartmentController + DepartmentHierarchyService
- LocationController + LocationStructureService
- DoorController + AccessPointStructureService
- TenantScopedModel provides ULIDs, soft deletion and scope helpers.

API conventions

- Route binding uses public_id, never numeric id.
- tenant_id and company_id come from trusted context, never client claims.
- PATCH writes go through services and audit listeners.
- 422 exposes invariant failures; 404 prevents cross-tenant disclosure.

Hierarchy-specific service rules

Service	Owns	Failure prevented
DepartmentHierarchyService	Create, move, archive and closure rebuild	Cycles, stale ancestor paths, orphaned descendants
LocationStructureService	Plant/branch exclusivity and coordinates	Ambiguous site ownership and partial geocoordinates
AccessPointStructureService	Gate/door scope and fail mode	Cross-location access points and unsafe silent changes

Verification and completion criteria

Phase 2 is complete when migration, SQL, models, services, API rules and diagrams describe the same organization boundary.

**Canonical migration**

17 tables created in dependency order and dropped in reverse order.

**Raw MySQL parity**

Generated DDL has the same table and column set as the Laravel migration.

**Tenant integrity**

Composite foreign keys and tenant-aware indexes reject cross-scope links.

**Department correctness**

Closure-table rebuild, subtree traversal and cycle rejection are specified and tested.

**Deletion safety**

RESTRICT preserves referenced history; service rules control archive operations.

**Static validation**

17 tables · 143 named keys/constraints · 17 models passed package checks.

Security and performance baseline

Security

- Trusted tenant context only.
- Policies enforce tenant/company/location authorization.
- Privileged safety changes require actor and reason.
- Core credentials and secrets never live in JSON settings.

Performance

- Organization indexes begin with tenant_id and usually company_id.
- No partitioning for low-volume Phase 2 masters.
- EXPLAIN ANALYZE required for location, subtree and door-by-gate queries.
- Monthly partitions begin only with immutable event tables.

Next phase

Phase 3 · Authentication and authorization: users, roles, permissions, scoped access, API clients, tokens and device credentials.